SHRIPAD VILASRAO DESHMUKH

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BACKGROUND

- **Reinforcement Learning** -- *RL Interpretability, Safe Exploration, Inverse RL, Agent-Based Models, Learning to Adapt*
- Document Analysis -- *Snippet Similarity, Multimodal Representation Learning, Object Detection, Semantic Segmentation*

EDUCATION

Indian Institute of Technology, Madras, Chennai, India *Aug'16 – Jul'20*
B.Tech. in Electrical Engineering, CGPA: *9.20/10*

PUBLICATIONS

- **S Deshmukh**, A Dasgupta, C Agarwal, N Jiang, B Krishnamurthy, G Theocharous, J Subramanian. "Explaining RL Decisions using Trajectories". *Accepted at International Conference on Learning Representations 2023.*
- **S Deshmukh***, A Java*, M Aggarwal, S Jandial, M Sarkar, B Krishnamurthy. "One-Shot Doc Snippet Detection: Powering Search in Document Beyond Text". *Published at IEEE/CVF Winter Conference on Applications of Computer Vision, 2023.*
- P Gupta, N Puri, S Verma, D Kayastha, **S Deshmukh**, B Krishnamurthy, S Singh. "Explain Your Move: Understanding Agent Actions Using Focused Feature Saliency". *Published at International Conference on Learning Representations, 2020.*

PATENTS

- S Singh, **S Deshmukh**, A Java, M Aggarwal, S Jandial, M Sarkar, A Jain, B Krishnamurthy. "FormProp: Novel method to propagate personalized error corrections across Forms corpus". *(Submitted for filing review)*
- A Java, **S Deshmukh**, M Aggarwal, S Jandial, M Sarkar, A Jain, B Krishnamurthy. "Forms Similarity Matching Framework for Enhancing RnC tool in AEM Forms". *(Approved for filing)*
- **S Deshmukh**, A Dasgupta, C Agarwal, B Krishnamurthy, G Theocharous, J Subramanian. "Novel Trajectory-based Explainability Framework for RL-based Decision Making". *(Submitted for filing review)*
- **S Deshmukh**, A Java, M Aggarwal, S Jandial, M Sarkar, A Jain, B Krishnamurthy. "A Novel Multimodal One-Shot Detection Approach for Document Snippet Search". *(Approved for filing)*
- S Jandial, **S Deshmukh**, B Krishnamurthy. "SelfDisturb: Novel Semantic Noise based Soft Label Regularization for Distilling Model Knowledge". *(Approved for filing)*
- **S Deshmukh**, S Jandial, A Java, M Aggarwal, M Sarkar, A Jain, B Krishnamurthy. "AdaData: Novel method to simplify data points for easier understanding of neural networks". *Attorney Docket No. P11364-US, 2022. (Filed)*
- S Jandial, S Ramesh, **S Deshmukh**, B Krishnamurthy. "Novel Method and Apparatus to Control Diffusion Model Image Generation". *Attorney Docket No. P11343 -US, 2022. (Filed)*
- **S Deshmukh**, M Aggarwal, M Sarkar, H Gupta. "Refining Element Associations for Form Structure Extraction". *Attorney Docket No. P10768-US, 2021. (Filed)*
- **S Deshmukh**, S Verma, J Subramanian, P Gupta, N Puri. "Customer Journey Management Using Machine Learning". *Attorney Docket No. P10405-US, 2021. (Filed)*

WORK EXPERIENCE

Machine Learning Research Associate, Adobe, Noida *Feb'22– Present*

(RL) Explaining RL Decisions using Trajectories. *In collaboration with G Theocharous & Prof. N Jiang (UIUC)* *Feb'22– Sept'22*

- Proposed a new explainability framework for RL that aims to attribute actions of RL agent to trajectories encountered during training. Introduced a Shapley value-based framework for performing such trajectory attribution.
- Introduced a single representation for an RL trajectory using trajectory transformers and demonstrated the utility of these embeddings in capturing semantically meaningful high-level behaviors.
- Qualitative and quantitative results on Grid-worlds, Atari & MuJoCo show the utility of our output explanations. Work **accepted at NeurIPS Workshop on Offline RL 2022.** The extended version is under review at another top-tier conference.

Novel Snippet-based Document Search Paradigm. *In Collaboration with M Sarkar* *Apr'22– Aug'22*

- To cater to the exigencies of sophisticated document search utilities, we proposed a novel one-shot document snippet detection task – a task to identify snippets similar to queried document snippet
- Conceptualized similarity metric for documents that can programmatically generate huge query-target dataset required for weak supervision of deep NNs. The metric circumvented issues of cost & high subjectivity in the human annotation.
- Co-designed multimodal cross-attention-based architecture, *MONOMER*, which beats state-of-the-art one-shot object detection & document analysis models. The work is **set to appear at WACV 2023.**

(RL) Agent-Based Models (ABMs) for Simulating Market Competition. *In Collaboration with J Subramanian* *Mar'22– Jun'22*

- Implemented multi-agent simulation of the document-sign market to capture the interaction among product companies & users. Programmed the simulation from scratch, referring to the COVID-19 ABM frameworks.

- Researched particulars about product strategies & user behaviors through consultancy reports. The simulation enabled a systematic study of emergent phenomena as market conditions are varied, hitherto speculated through human intuition.

Machine Learning Engineer, Adobe, Noida *(Work from Home)* *Jun'21 – Jan'22*

Shipment of 'List' and 'Section' detection models to Adobe's Automated Forms Conversion Service *Dec'21 – Jan'22*

- Deployed *Deformable DETR*-based list & section detection models leading to ~11% gain in class conversion quality

Conversion of Flamingo Forms Dataset to MS-COCO format *Sept'21 – Nov'21*

- Came up with an idea to convert *Flamingo Forms*, an Adobe proprietary document dataset, to standard *MS-COCO* format, which enabled us to leverage the latest developments in ever-growing image recognition fields
- Benchmarked performance of over top 10 semantic segmentation + object detection approaches on 18 document classes. This exercise **boosted the average conversion quality of the entire service by 8%.**

Detecting Table Bounds in Form Documents *May'21 – Jul'21*

- Trained Semantic Segmentation models '*EfficientDet*', '*RetinaNet*', and Object Detection approaches '*YOLOv3*' & '*YOLOv5*' to capture bounds of tables within *blank* forms. With an improvement of over 9.5%, shipped the best model.

Member of Technical Staff, Adobe, Noida *(Work from Home)* *Aug'20 – May'21*

(RL) Risk minimization in Customer Journey Management using Inverse RL *Oct'20 – Dec'20*

- Ideated use of trajectory-based inverse RL methods for minimizing risk involved while intervening in a customer journey.
- Empirically demonstrated more than 20% reduction in customer churn rate. **Filed patent for the idea**.

Multimodal Element Association *Oct'20 – Feb'21*

- Debugged and deployed *TensorFlow 1.x* codebase for using segmentation masks as prior in a multimodal element association pipeline. This resulted in 15% precision and 4% recall gain in extracting higher-order document structures.
- Drafted patent-idea for prior inclusion, '*Refining Element Association…*' which is now **part of exemplary drafts** on *Adobe Brightidea* (the only new addition to the list in 2021).

INTERNSHIPS

Summer Research Intern, Adobe, Noida *May'19 – Aug'19*

(RL) Explaining Reinforcement Learning Agents. *Mentors: N Puri, S Verma, P Gupta*

- Collaboratively proposed a new saliency metric for visual explanations of RL policy, significantly outperforming prior art in terms of specificity and relevance of the generated saliency maps.
- Co-Produced explanations for *Stockfish* chess engine. Extended the idea to *LeelaChess-0*, *Minigo* Go agent, and Atari game-agents trained for *Pong*, *Breakout,* and *Space-Invaders*. **Published at ICLR 2020.**

Research & Development (R&D) Intern, Center for Development of Advanced Computing (CDAC), Pune *Dec'18 – Jan '19*

Matrix and Graph Computations in High-Performance Computing. *Mentors: Prof. N Karmarkar, Dr. VCV Rao*

- Part of National Supercomputing Misson, a project aimed at creating a supercomputer cluster to empower Indian R&D
- Developed FORTRAN libraries to test & extract the performance of Intel Linear Algebra Kernels for matrices of varying specs
- Successfully integrated all the existent modules developed for the project under one Unified Object-Oriented Interface, along with complete documentation. Additionally, worked on algorithms for generating hyper-graphs.

SCHOLASTIC AWARDS & ACHIEVEMENTS

- National Prize Certificate holder for top students at IIT Madras (1 out of 5 recipients in a batch of 120 EE students) *2017*
- Secured All India Rank (AIR) 323 in JEE Advanced (top 0.16%) and AIR 491 in JEE Mains (top 0.03%) *2016*
- Selected for Kishore Vaigyanik Protsahan Yojana Fellowship with an overall national rank of 621 *2015*
- Ranked among the top 1% in the National Chemistry Olympiad conducted by Homi Bhabha Institute for Science Edu. *2015*
- Recipient of National Talent Search Exam (NTSE) Scholarship (Stage 1 (of 2) – Maharashtra State Topper) *2012*

RELEVANT COURSEWORK

Reinforcement Learning	Dynamic Games: Theory & Applications	Pattern Recognition & Machine Learning
Convex Optimization	Linear Algebra Techniques for Data Analysis	Probability, Statistics & Stochastic Process

MENTORSHIP & VOLUNTEERING EXPERIENCE

Mentor at Adobe, Noida *Summer'22*

- Mentored **three** research interns working on various problems in the field of Reinforcement Learning

Moderator at Liberty, Equality, and Fraternity (LEAF) Society, IIT Madras *Jan'20 – May'20*

- Founding member of LEAF which is a society of 20+ students established to instill a culture of healthy discussions on diverse socio-economic issues faced by the world, and India in particular.
- Got elected as a moderator for the Jan-May'20 semester. Conducted 8 discussion sessions during my tenure. Sessions included discussion on topics such as agrarian distress, the COVID-19 situation, debate on human rights, etc.

Coordinator at Extra Mural Lectures (EML), IIT Madras *Jun'17 – May'18*

- EML is an official IIT Madras student body that organizes lectures on various topics: Arts, Technology, Culture & the like.
- As a part of EML, got an opportunity to organize lectures by luminaries such as Orkut Büyükkökten (Founder of Orkut), Nirmala Sitharaman (then Defence Minister), Dr. Prakash Amte (Magsaysay Award Winner), D. Subbarao (Ex. RBI Governor).
- I also organized a lecture at SRM university, outside the IITM, as part of the initiative to expand EML nationwide.